



BITS Pilani
Pilani Campus

Module 2: Mechanical Separations (Chapter 29)

Lecture 4

Somak Chatterjee

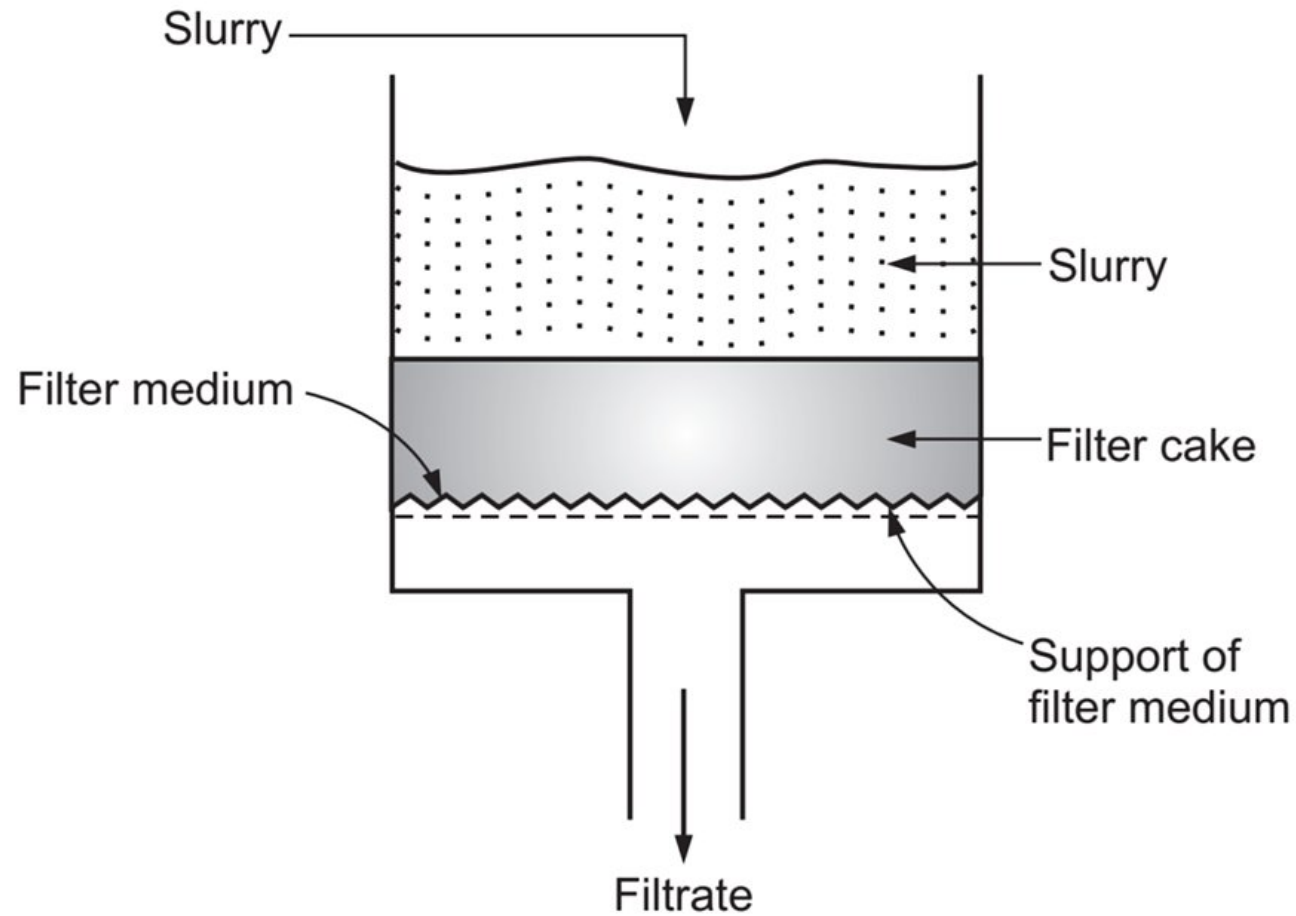


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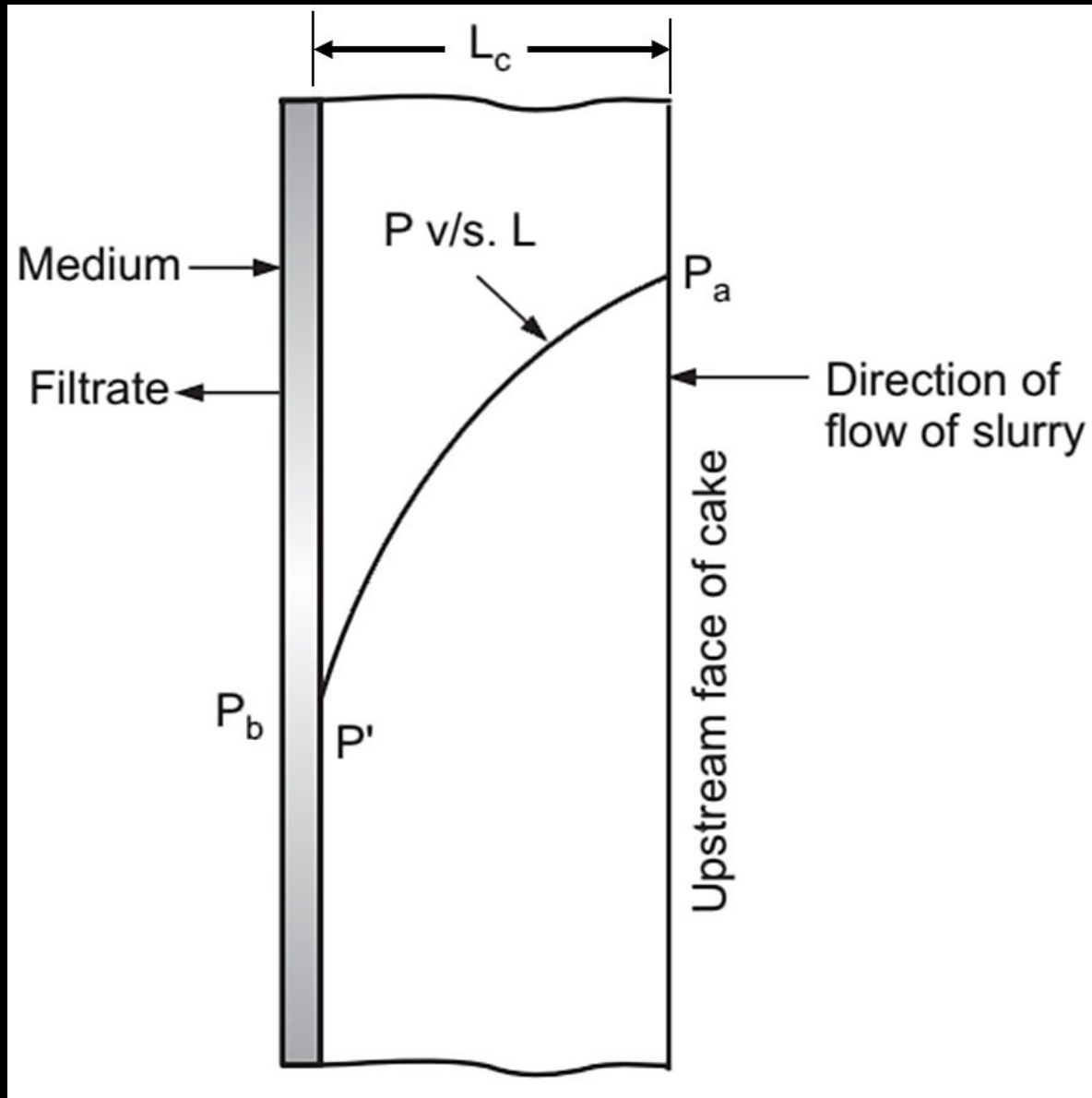
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Lecture: Constant Rate Filtration

Principles of Cake Filtration



PRESSURE DROP THROUGH FILTER CAKE



Principles of Cake Filtration



- **Constant-pressure filtration:**

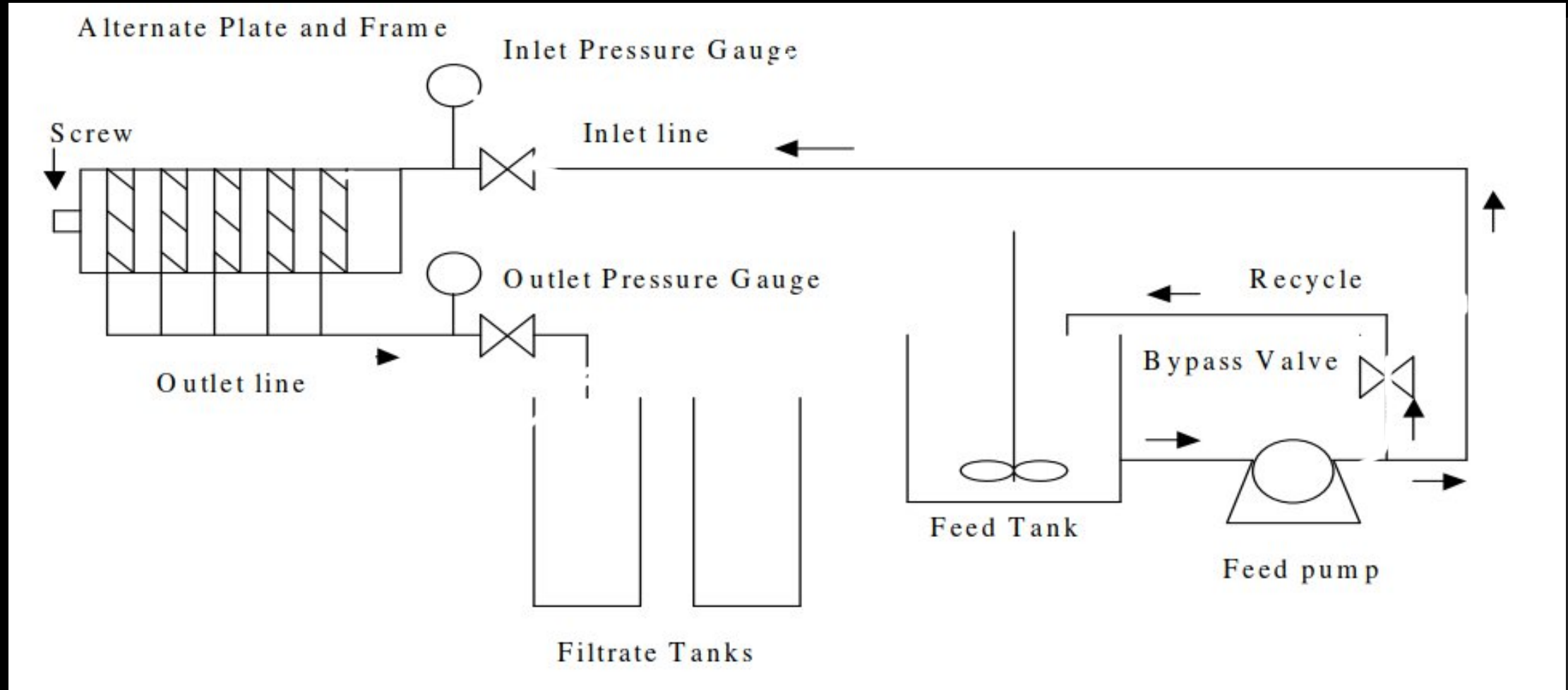
- The method in which the pressure drop over the filter is held constant through the run so that the rate of filtration is maximum at the start of filtration and decreases continuously towards the end of the run

- **Constant-rate filtration:**

- The method in which the pressure drop is varied usually from a minimum at the start of filtration to a maximum at the end of filtration.

- **Continuous filtration:**
 - In a continuous filter such as rotary drum filter, the feed, filtrate and cake move at steady constant rates.
 - But for any particular element of the filter surface, conditions are unsteady.
 - The process of filtration consists of several steps in series such as cake formation, washing, drying and discharging/scrapping and each of these steps involve gradual and continual change in conditions but the pressure drop across the filter during cake formation is held constant.
 - Therefore, the equations obtained for discontinuous constant-pressure filtration may be applied to continuous filters with some changes.

Plate and Frame filter press



Lecture 10

Constant rate filtration



Filtrate flowing at constant rate.

u (linear velocity) = Constant

$$u = \frac{1}{A} \frac{dv}{dt} = \frac{V}{At}$$

$$\alpha = \frac{\Delta P_c g_c A}{\mu u m_c}$$

$$m_c = VC$$

$$\frac{\Delta P_c}{\alpha} = \frac{\mu}{g_c A} \frac{V}{At} VC$$

$$\frac{\Delta P}{\alpha} = \left(\frac{\mu c}{g_c t} \right) \left(\frac{V}{A} \right)^2 = \left(\frac{\mu c}{g_c t} \right) \left(\frac{V}{A} \right)^2$$

In the above Eqn,
 $\frac{\Delta P}{\alpha}$ is retained on left
hand side, as $d = f(\Delta P)$
for Compressible Sludges.

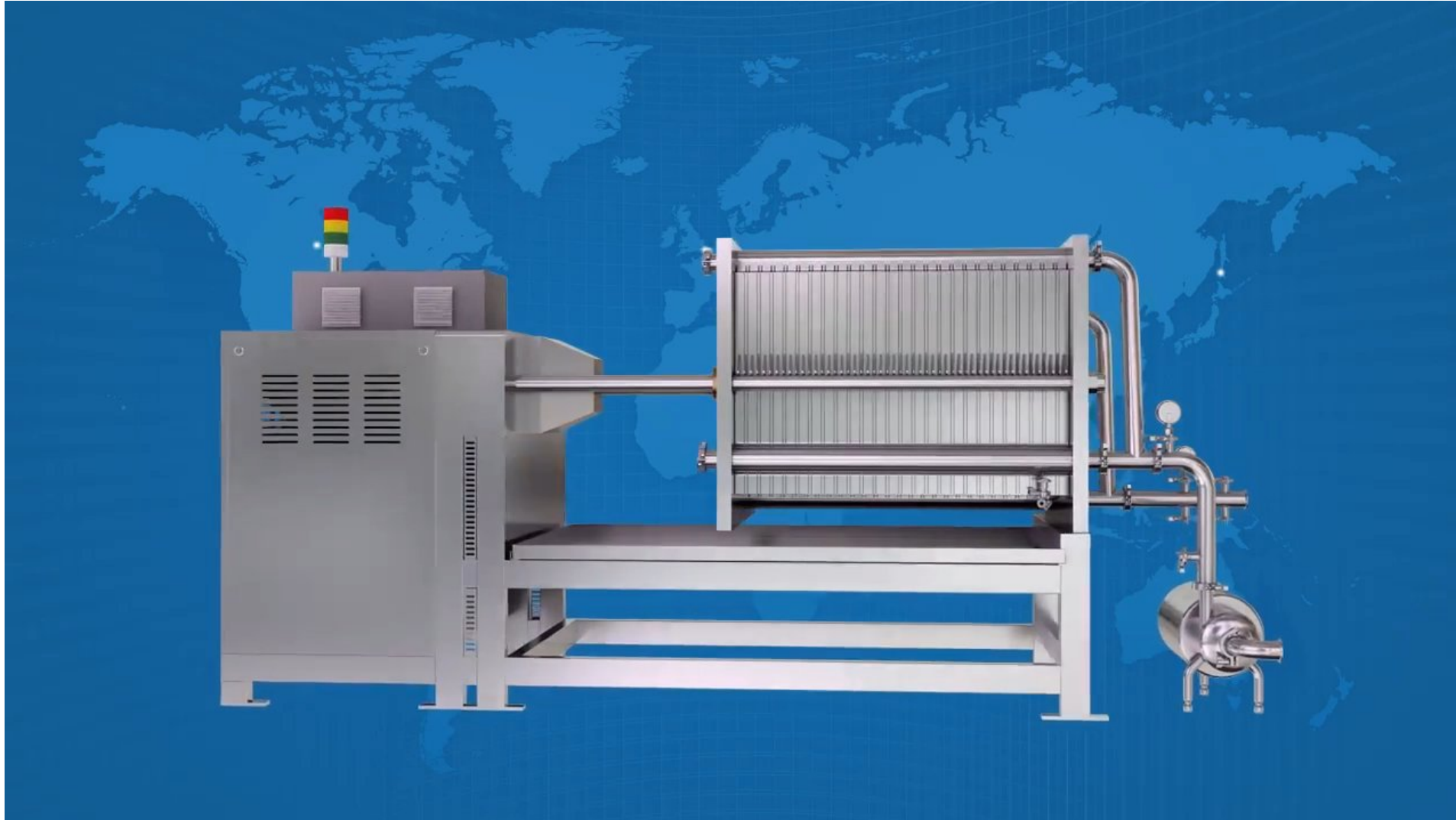
Plate and Frame filter press

innovate

achieve

lead

<https://www.youtube.com/watch?v=IOwnTwUYNAg>



References



- Smith, J., Harriott, P., McCabe, W. (2005). Unit Operations of Chemical Engineering. Colombia: McGraw-Hill Education.
- Swain, A. K., Patra, H., & Roy, G. K. (2011). Mechanical operations. Tata McGraw Hill.